Postoperative Recurrence Ascertainment Under Intermittent Surveillance and Competing Mortality

ChatGPT

Abstract

Background. Recurrence after pancreatic cancer resection may precede its clinical documentation. Death between examinations can prevent recurrence from being recorded.

Methods. We distinguish the onset of clinically detectable recurrence from its documentation at surveillance visits. We derive exact ascertainment probabilities in an illness–death model and assumption-dependent nonparametric bounds, with finite-sample confidence limits. We evaluate irregular observation and death through simulation and summarize published recurrence counts from the ESPAC-4 trial.

Results. Six-month surveillance documented 43.7% of patients by two years in a synthetic scenario with a 53.5% underlying recurrence risk. The direction of distortion from treating death as censoring varied relative to underlying recurrence risk. An exact confidence construction covered the target under perfect final examinations; a deliberate false-negative scenario invalidated the bounds.

Conclusions. Documented and underlying recurrence are distinct clinical estimands. Their interpretation depends on examination completeness, diagnostic accuracy, and mortality before confirmation. Published trial aggregates are informative about recurrence burden but insufficient to estimate the unobserved two-year recurrence process.

1 Introduction

Resection offers some patients with pancreatic ductal adenocarcinoma (PDAC) an opportunity for prolonged survival, but disease recurrence is frequent. In a secondary analysis of the ESPAC-4 trial, recurrence was documented in 479 of 730 participants, and 78 died without documented recurrence [Jones et al., 2019]. These categories describe recorded clinical histories, not necessarily the time at which recurrent disease first became detectable.

Postoperative examinations are intermittent. Scan schedules vary, and distinguishing recurrence from postoperative changes is difficult [Chu et al., 2023]. Surveillance policy itself is associated with the detection and management of recurrence in observational clinical studies [Andel et al., 2025]. A patient can develop recurrence after a negative examination and die

before the next scheduled scan. Recording the death as a competing event is necessary, but it does not retrospectively reveal whether recurrence occurred before death.

Interval-censored illness–death and semi-competing-risk models offer principled approaches to this setting [Xu et al., 2010, Thao et al., 2024, Morsomme et al., 2025]. Informative examination times are another established concern [Lange et al., 2015]. The present analysis does not claim to introduce these models. Rather, its objective is to quantify the difference between documented and underlying recurrence, state the assumptions required for interpretable bounds, and demonstrate the relevant failures under explicit violations.

2 Estimands and observation structure

Time zero is surgery. Define a three-state process: state 0, alive without clinically detectable recurrence; state 1, alive with clinically detectable recurrence; and state 2, dead. Let T_R denote the transition from state 0 to state 1, if it occurs before death, and let T_D be death time. A recurrence event is not defined after death. The primary clinical estimand is

$$F_R(H) = \mathbb{P}(T_R \leq H,\ T_R < T_D), \tag{1}$$

the probability of entering the detectable-recurrence state before death by horizon H. This does not mean the time of initial tumor-cell regrowth. The documented-recurrence estimand is

$$D_V(H) = \mathbb{P}(\text{a positive surveillance examination at or before } H), \tag{2}$$

where V denotes the examination schedule. The two quantities are not interchangeable.

We first assume examinations are perfect for detecting a persistent state 1; every living patient is examined at time H; exact death times are known; and positive results indicate true recurrence. The examination schedule may differ between patients, provided the final examination condition holds. We later relax the perfect-test assumption experimentally. An observation at H identifies current state among survivors, but death between visits can obscure an earlier state transition.

3 Ascertainment identities and bounds

3.1 Exact calculations in a simple illness–death model

Assume constant transition intensities a for $0 \to 1$, b for $0 \to 2$, and c for $1 \to 2$. For a schedule $0 = v_0 < v_1 < \cdots < v_K = H$, the probability density of recurrence at time u prior to death is $a \exp[-(a+b)u]$. Recurrence in $(v_{k-1}, v_k]$ is documented at v_k only if the patient

survives from u to v_k. Thus

$$F_R(H) = \frac{a}{a+b}\Big\{1 - e^{-(a+b)H}\Big\}, \qquad D_V(H) = \sum_{k=1}^{K} \int_{v_{k-1}}^{v_k} a e^{-(a+b)u} e^{-c(v_k-u)}\, du. \tag{3}$$

The continuous extension is used when $a + b = 0$. The expression describes documentation, not a likelihood fit to patient data.

Proposition 1 (Surveillance loss)**.** *Under the illness–death assumptions,* $0 \le D_V(H) \le F_R(H)$*. Adding examinations without deleting existing examinations cannot decrease* $D_V(H)$*. When* $c = 0$*, and all living participants are examined at* H*,* $D_V(H) = F_R(H)$*. For* $a > 0$*,* $c > 0$*, and a nondegenerate interval between exams, the inequality is strict.*

Proof. Each integrand of $D_V(H)$ equals the recurrence-onset density multiplied by the conditional probability of survival to the next examination, which lies in $[0, 1]$. Dropping this factor and summing over examination intervals gives $F_R(H)$. Under a refined schedule, every individual whose recurrence is observed under the original schedule is also observed under the new schedule: state 1 persists until death, and all original examination times remain available. When $c = 0$, survival in state 1 to the final exam is certain. If $a, c > 0$, there is positive probability of recurrence followed by death within an open examination interval, producing a positive difference. □

This identity isolates ascertainment loss due to death between examinations. It is not a causal effect of imaging frequency on survival or tumor biology. For a regular interval Δ dividing H, the integral has closed form

$$D_\Delta(H) = a \sum_{k=1}^{H/\Delta} e^{-(a+b)k\Delta} \frac{1 - e^{-[c-(a+b)]\Delta}}{c - (a+b)}, \tag{4}$$

with the quotient replaced by Δ if $c = a + b$.

Proposition 2 (Competing-bias reversal)**.** *Consider the regular-schedule illness–death model with* $a, b > 0$*, fixed* H *and at least one scheduled examination in* $(0, H]$*. Let* $K_V(H)$ *denote the large-sample limit of one minus the Kaplan–Meier estimator that treats death before documented recurrence as censoring. At* $c = 0$*,* $K_V(H) > F_R(H)$*; as* $c \to \infty$*,* $K_V(H) \to 0 < F_R(H)$*. Hence, for some intermediate values of* c*, the same invalid censoring convention may overestimate or underestimate underlying recurrence risk.*

Proof. The Kaplan–Meier complement for first documented recurrence equals one minus the product of conditional non-event probabilities at the examination times, with deaths removed from the risk sets. Its product differs from the cumulative-incidence calculation precisely because the competing deaths are removed rather than retained in the population denominator. At $c = 0$, every recurrence before H is eventually documented by the final

examination, so the cumulative incidence of documentation equals $F_R(H)$. Since $b > 0$, a positive mass of participants die in state 0 before some subsequent documented recurrence, and treating these deaths as censoring strictly raises the Kaplan–Meier complement above the documented cumulative incidence. As $c \to \infty$, a person entering state 1 survives to the next positive-length examination gap with probability tending to zero. Therefore each documented event mass tends to zero, while each corresponding risk-set probability remains bounded away from zero because $\mathbb{P}(X(t) = 0) > 0$ at finite t. Consequently $K_V(H) \to 0$. The underlying $F_R(H)$ does not depend on c and is positive. Continuity establishes the change in sign. □

3.2 A model-free interval with explicit assumptions

Let A denote documentation of recurrence at or before H, and B denote death before H without previously documented recurrence. The events A and B are disjoint. We require complete state ascertainment at H for every survivor, persistence of detectable recurrence, and perfect classification at observed examinations. For deaths without documented recurrence, assume the last negative examination occurred strictly before death, leaving an unexamined interval. No upper limit on recurrence intensity is imposed.

Proposition 3 (Sharp ascertainment interval)**.** *Under these assumptions, the observed-data distribution implies*

$$\mathbb{P}(A) \leq F_R(H) \leq \mathbb{P}(A) + \mathbb{P}(B). \tag{5}$$

These are sharp bounds in the unrestricted illness–death model conditional on the observed examination and death histories, apart from boundary cases with exact state ascertainment at death.

Proof. Every documented recurrence must have preceded death by H, so A is a subset of the target event. A participant who is alive with a negative examination at H cannot have experienced a persistent clinically detectable recurrence by H. Therefore any recurrence not belonging to A must concern someone who died without documentation, giving the upper inclusion in $A \cup B$. For sharpness, leave all documented histories unchanged. For any chosen subset of individuals in B, insert an unobserved transition to state 1 strictly between that individual's final negative examination and death; for the remainder, keep the person in state 0 until death. The observed examination results and exact death times are unchanged. Assigning zero, all, or arbitrary fractions of B to the unobserved transition produces every risk between the endpoints, allowing mixture distributions if necessary. This construction relies on the absence of restrictions on transition hazards and no definitive state assessment at death. □

If n independent participants have identical recruitment criteria and follow-up to H, let

$\widehat{p}_A = n^{-1}\sum_i \mathbf{1}(A_i)$ and $\widehat{p}_{A\cup B} = n^{-1}\sum_i \mathbf{1}(A_i \cup B_i)$. For $0 < \alpha < 1$, set

$$r_n = \sqrt{\frac{\log(4/\alpha)}{2n}}, \quad C_n = \left[\max(0, \widehat{p}_A - r_n), \min(1, \widehat{p}_{A\cup B} + r_n)\right]. \tag{6}$$

Proposition 4 (Finite-sample confidence set)**.** *Under the conditions of Proposition 3,* $\mathbb{P}\{F_R(H) \in C_n\} \geq 1 - \alpha$ *for every sample size* n*.*

Proof. Both indicators $\mathbf{1}(A_i)$ and $\mathbf{1}(A_i \cup B_i)$ are bounded independent variables. Hoeffding's inequality and a union bound show that both respective expectations differ from their empirical means by at most r_n with probability at least $1 - 4\exp(-2nr_n^2) = 1 - \alpha$. On this event, both inequalities in (5) imply that the target lies in C_n. □

Under the constant-rate model, the upper endpoint has a particularly simple form:

$$\mathbb{P}(A) + \mathbb{P}(B) = 1 - \mathbb{P}\{X(H) = 0\} = 1 - e^{-(a+b)H}. \tag{7}$$

Indeed, with perfect final examinations, the only participants outside $A \cup B$ are living and in state 0 at H. Hence the upper endpoint is unaffected by examination frequency or post-recurrence death intensity c, even though the lower endpoint changes. This is a feature of the weak, unrestricted bound, not evidence that surveillance timing is immaterial.

The confidence set is intentionally conservative and describes a population with the stated follow-up protocol. It must not be applied to aggregate recurrence counts when examination timing or completeness is unknown. Imperfect sensitivity among survivors can invalidate the upper bound because a negative final examination need not exclude recurrence.

4 Numerical investigation

We simulated $n = 1500$ independent postoperative histories per replication over $H = 2$ years, with $a = 0.43$ and $b = 0.09$ per year except for one deliberate assumption-violation scenario. Death after recurrence was varied through c, and examination intervals were three or six months. Each scenario had 500 independently generated cohorts. For comparison, we calculated the crude documented fraction and a Kaplan–Meier estimator that incorrectly treats deaths before documentation as noninformative censoring. The displayed theoretical recurrence risk refers to (1); the recorded fraction estimates (2).

The death-censored Kaplan–Meier estimate was greater than the documented fraction in every scenario. It exceeded the underlying recurrence risk in the low-mortality setting but fell below it under higher mortality or less frequent surveillance. These opposing biases can partially offset one another numerically, which does not make death censoring valid. With perfect tests and final examinations, the finite-sample confidence set (6) covered the true risk in all 500 replications in each of the five correctly specified scenarios. This is empirical coverage, not a substitute for Proposition 4.

Table 1: Monte Carlo analysis of recurrence ascertainment over two years (500 replications per scenario). All percentages are synthetic.

Scenario	True risk	Documented	Death-censored KM	Bound upper
3-month, low post-recurrence mortality	53.5	53.1	57.9	64.6
3-month, moderate mortality	53.5	48.3	54.6	64.6
3-month, high mortality	53.5	43.9	51.4	64.7
6-month, moderate mortality	53.5	43.7	51.9	64.7
6-month, high mortality	53.5	36.2	45.9	64.7

Entries are mean estimated percentages except true risk, calculated analytically. The upper endpoint is $\widehat{p}_{A\cup B}$, not the confidence limit. Post-recurrence death rates were 0.05, 0.8 and 1.6 per year for low, moderate and high scenarios.

To test the assumptions, we reduced examination sensitivity to 40%, set $b = 0.02$ and $c = 0.05$ per year, and retained six-month examinations. Across 500 replications, the actual two-year recurrence risk was 56.7%, but the mean observed upper endpoint was only 43.9%. The putative confidence sets covered the true risk in none of these replications. Thus the final-examination accuracy assumption is consequential rather than cosmetic.

5 Illustration using published clinical data

The ESPAC-4 recurrence analysis reported 479 of 730 participants with documented recurrence, 78 deaths without recorded recurrence, and 173 participants alive without documented recurrence at the study's reported assessment [Jones et al., 2019]. Of the 479 recurrences, 238 were local only, 193 distant only, and 48 local and distant concurrently. These are published counts, not a reconstruction of patient-level event times.

Table 2: Descriptive proportions from the published ESPAC-4 recurrence analysis.

Published category	Count	Cohort percentage
Documented recurrence	479	65.6
Death without documented recurrence	78	10.7
Alive without documented recurrence	173	23.7

As an illustration of classification sensitivity, suppose that a fraction q of the 78 deaths without documented recurrence had previously experienced an unobserved recurrence. Reclassifying only these cases would change the descriptive recurrence fraction to $(479 + 78q)/730$, from 65.6% at $q = 0$ to 76.3% at $q = 1$. This is a hypothetical reassignment, neither an identified interval for underlying recurrence nor an estimate of two-year risk. The available aggregates do not establish that the 173 living patients had definitive negative examinations at a common horizon, and their follow-up times differ. Therefore Propositions 3–4 are not

applied to these counts. Individual operation dates, examination dates, test results, and death records would be required.

6 Discussion

Our results distinguish three questions. The clinical process determines when recurrent disease becomes detectable. Surveillance determines whether that event is documented before death. Mortality determines whether another opportunity to document recurrence remains. Estimates formed from the recorded first-positive dates principally characterize the second question unless the observation model is explicitly incorporated.

The nonparametric interval is interpretable but may be wide when mortality before documentation is frequent. Its strength is transparency: it does not assume proportional hazards, independent examination times, or a particular transition intensity. Its weakness is equally important: perfect examinations and complete final follow-up are strong assumptions in postoperative PDAC. Imaging uncertainty, missed appointments, differential surveillance, and treatment-induced changes can all weaken the direct interpretation of a negative scan [Chu et al., 2023]. Our false-negative experiment illustrates the possible consequence. Fully specified multistate models can address more complex clinical records, but are already well developed in the literature [Xu et al., 2010, Thao et al., 2024, Morsomme et al., 2025].

The clinical illustration is deliberately restricted to published ESPAC-4 category counts. It cannot estimate an unobserved recurrence hazard, test the examination model, or quantify a two-year postoperative recurrence risk. Likewise, the simulations are hypothetical and do not estimate the efficacy of more frequent imaging or its effect on survival. Future clinical work requires dated postoperative surveillance records and careful validation of image-based recurrence classification.

The principal contribution of this draft is therefore an explicit ascertainment analysis and reproducible comparison of clinical estimands, rather than a claim of a new general survival-analysis theorem or prospective clinical validation.

Data, code, and AI-generation declaration

The clinical counts were transcribed from the published ESPAC-4 secondary analysis and are reproduced in the accompanying Python script with their source identified. No identifiable patient records were obtained. All simulated data and numerical results were generated by Python code included in the research package. This manuscript, mathematical derivations, code, and writing were fully generated by artificial intelligence and may contain errors. They require independent mathematical, computational, clinical, and bibliographic verification before any submission or medical use. No clinical data collection or human-subject intervention was undertaken for this draft.

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